The Role of Human Agency in Human-AI Co-Creativity

Sarah H. Wu
Stanford University
Stanford, CA, USA

Yuewen Yang
Cornell University
Ithaca, New York, USA

Angela Y. Lee
University of Wisconsin-Madison
Madison, Wisconsin, USA

Alex Liebscher
BetterUp
Austin, Texas, USA

Kristina Rapuano
BetterUp
Austin, Texas, USA

Kate Niederhoffer
BetterUp
Austin, Texas, USA

Jeffrey T. Hancock
Stanford University
Stanford, California, USA

Abstract

Large language models (LLMs) are popular tools for creative ideation, but have been shown to homogenize outputs across users. We test if approaching an AI tool with high human (vs. low) human agency can mitigate this homogenization effect by encouraging people to use AI to augment their creativity, rather than offload it. Participants were experimentally assigned to one of three conditions (high-agency approach with AI access, low-agency approach with AI access, or a human-only control) and generated creative uses for everyday objects. Contrary to our expectations, ideas did not differ in individual-level quality (overall creativity, originality, and usefulness). However, a preregistered similarity-to-centroid analysis and an exploratory cluster analysis provided convergent evidence of AI-induced homogenization among the low-agency condition. Thus, while AI has enabled greater ideational fluency, our research suggests the degree of agency with which people approach their AI tool has downstream consequences on collective creativity.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; *Empirical studies in HCI*; *Ubiquitous and mobile computing systems and tools.*

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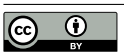
1 Introduction

One of the cornerstones of human cognition and communication is our ability to engage in creative ideation: the process of generating, developing and conveying new ideas. People have long relied on tools to support their creative processes, from physical notebooks to mind-mapping software [6]. One of the newest and most popular tools to stimulate creativity are large language models (LLMs), which can provide a thought partner to anyone with an internet connection. Fundamentally, LLMs are token predictors that output likely words given a user prompt and context. Their ability to rapidly synthesize massive amounts of text make them useful for a wide range of creativity tasks, and recent reports suggest that people regularly use these systems for such purposes [15].

The process of people using LLM systems for ideating and communicating creative ideas we refer to as *human-AI co-creativity*. Research to date suggests two important patterns in human-AI co-creativity. The first is that LLM systems perform as well as or even better than humans on commonly accepted creative ideation benchmarks [8] and benchmarks for ideation in specific contexts [7]. However, when humans *use* AI systems to be creative, there is a risk of homogenization: People who use AI tend to produce more similar outputs compared to their counterparts who do not—an effect that has been shown in a range of contexts involving open-ended output, such as product design [1, 18] and short-story writing [4]. In recent years computer scientists have termed this paradox *algorithmic monoculture* [10], or, in the case of generative AI, *generative monoculture* [25]. Derived from the notion of monoculture in agriculture, these terms capture the vulnerabilities incurred by mass reliance on a single algorithm. While each AI-assisted individual may generate an idea more creative than what they could achieve on their own, this comes at a risk on the collective level. What is creative with respect to an individual may in fact be trite among a group of individuals using the same AI assistant, especially if many individuals use AI without bringing their own human experience to the interaction.

In the present study, we examine the extent to which human agency impacts the homogenizing effects of LLMs or preserves the benefits of human-AI co-creativity. Specifically, we explore whether a high-agency approach, in which a person approaches a tool in a way that supports their own agency (rather than replacing it) has downstream consequences on individual- and collective-level ideation outcomes. For example, when people exercise high agency over technologies such as social media, they believe they can control the amount and nature of their use [12]. Similarly, encouraging individuals to develop a more intentional orientation towards their AI use may lead them to engage more in creative ideation and increase their likelihood of contributing diverse output to a group.

We test the importance of a high-agency approach in human-AI co-creativity by experimentally manipulating the guidance people



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receive about AI as a creativity support tool. We compare a group of people who are prohibited from using AI against two treatment groups that were told to use AI: a low-agency condition, in which participants are instructed to let AI do the bulk of the creative ideation, and a high-agency condition, in which participants are instructed to leverage their own human creativity in tandem with the AI to develop even better ideas. We assess creativity at both the individual and group levels to explore the effect of human agency on both the quality and diversity of ideas produced in human-AI collaboration.

2 Method

2.1 Participants

We recruited two samples of adult participants from the U.S. to participate in the study: a sample of submitters and a sample of evaluators. Submitters were community college students in the San Francisco Bay Area (N = 150; age: M = 25.1 years, SD = 9.2; 72% female, 24% male, 1.3% non-binary, 2.7% self-described) who participated for research credits. An additional three submitters were excluded for failing an initial attention check. Evaluators were U.S. adults recruited from the CloudResearch Connect platform [9] (N = 142; age: M = 46.7 years, SD = 16.3; 49% female, 51% male) who were compensated financially for their time. Six participants were excluded for failing an initial attention check and five participants were excluded due to idea data logging issues.

2.2 Procedure

Methods were approved by the Stanford Institutional Review Board. All participants provided informed consent.

2.2.1 Submission Phase. Upon consent, participants were asked to read a general description of the Alternative Uses Task (AUT), in which they were asked to generate non-standard uses for three ordinary objects: bubble wrap, frisbee and brick. They were provided with a rubric with our definitions of overall creativity and its two canonical subcomponents: originality and usefulnessb[22]. After consenting, participants were randomly assigned to one of three conditions that varied whether they completed the idea-generation task with or without access to an AI chatbot (please see Figure 1 for representative excerpts from the instructions shown in each condition). The low- and high-agency conditions provided participants with access to a custom AI chatbot that was designed for the study, which was built on top of gpt-3.5 and embedded into the survey through a web-application. Participants in the low-agency condition were told that "AI recognizes patterns, styles, and connections that would take years of study to master on your own…The beauty of AI is that it does the heavy lifting for you." They were instructed to let AI's creative power work for them. Conversely, in the high-agency condition, participants were instructed to use AI to expand upon their own ideas and generate ones that are even more creative. This aspect of the manipulation was informed by prior research on growth mindsets, in which interventions encourage strengthening one's cognitive muscles [26].

In the control condition, participants completed the same AUT without access to an AI chatbot. Given that past work suggests that online survey respondents frequently use AI tools to generate answers [24], we explicitly instructed participants in the control condition to not use AI in this task. Before the start of the AUT, participants read, "We value your individual human creativity, so please do NOT use AI during the task." To reduce spillover, copy-paste functionality was disabled using JavaScript code embedded in the Qualtrics survey. Participants were required to spend a minimum of 10 seconds on the instruction page in each condition before proceeding to the AUT.

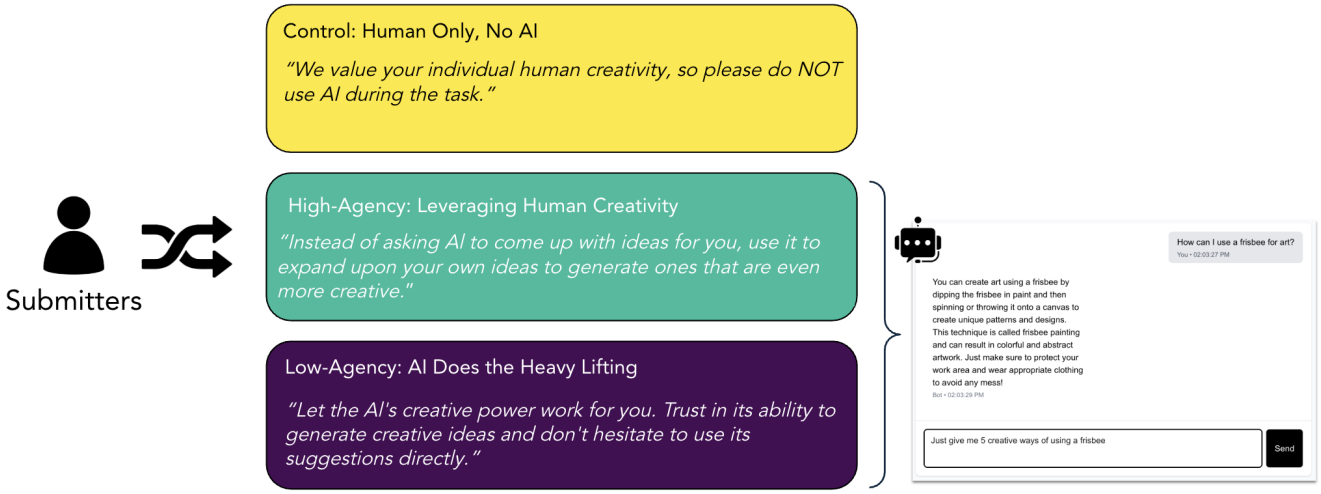
For each of three objects (frisbee, bubble wrap, and brick), submitters were asked to generate five alternative uses for the object and enter their ideas in form fields. The presentation of the three objects was counterbalanced. Participants in the high-agency and low-agency conditions had access to ChatGPT through an embedded web-app in the Qualtrics survey; they entered their alternative uses below the web-app interface. Participants in the control condition had no web-app, just the fields to enter their alternative uses for each object. In line with prior research on induction with the AUT [14], all participants were given as much time as they needed to complete the task. This avoided penalizing the participants in the low-agency and high-agency conditions for taking time to work with the AI chatbot. Across conditions, submitters produced a total of n = 2,250 ideas.

Finally, all submitters provided a brief description of their idea generation process in a text box. Participants in the high-agency and low-agency conditions were asked to self-report the extent to which the AI helped them with the creative process (1 = Strongly Disagree to 7 = Strongly Agree; [13], the extent to which they perceived AI as a conversational partner (1 = Strongly Disagree to 7 = Strongly Agree; ω = 0.86) [13], and their intrinsic motivation during the AUT (how interesting, enjoyable, fun, and engaging it was; 1 = Not at all to 7 = Completely; ω = 0.92) [23]. All participants answered demographic questions before being debriefed on the purpose of the study and granted research credits for their participation.

2.2.2 Evaluation Phase. Next, we asked our evaluators to rate the compliant ideas from the submitters. Evaluators were shown the same definitions of creativity, originality, and usefulness as the submitters, which served as their rubric upon which to score the ideas. Each evaluator rated a random subset of 15 of the total set of ideas. The submitted ideas were stratified by object type, such that each evaluator rated five ideas for bubble wrap, five ideas for a brick, and five ideas for a frisbee. For each of the 15 ideas they saw, the evaluators rated it in terms of overall creativity, originality, and usefulness, resulting in 45 ratings produced by each evaluator. Due to our randomization procedure, some ideas were evaluated multiple times, while others were never evaluated and thus were excluded from this analysis. All participants answered demographic questions before being debriefed on the purpose of the study and compensated for their participation.

2.2.3 Manipulation Checks. We conducted manipulation checks to ensure that submitters followed the instructions for their conditions. Using the chat log data, we identified trials in which participants in the high-agency and low-agency treatment conditions did not use the AI tool as instructed. In 24% of low-agency trials and 38% of high-agency trials participants did not use the AI tool. Object trials from the high- and low-agency conditions were excluded if they did not include chat data. On the participant level this resulted in

Figure 1: Experimental Paradigm

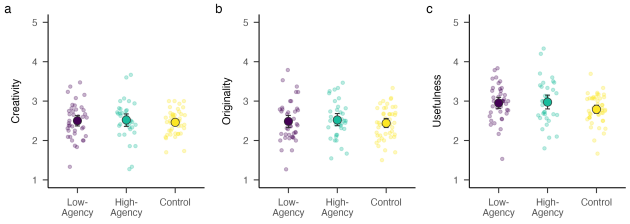


Note: Submitters were randomly assigned to one of three between-subjects conditions, in which they read one of three sets of instructions before completing the Alternative Uses Task. The key excerpt from the interventions are shown in the figure. In the high- and low-agency conditions, participants interacted with an AI chatbot to develop their creative ideas; chat data was logged on the backend.

12 participants in the high-agency condition and 10 participants in the low-agency condition being completely excluded from the analysis.

2.2.4 Analytic Approach. We assessed the creative qualities of participants' submitted ideas at both the individual level and the condition level. To assess idea quality, we examined the ratings from evaluators at the individual participant level. To assess homogeneity, we first used the sentence embedding model all-MiniLM-L6-v2 [20] to generate embeddings in a 384-dimension embedding space. Using these embeddings, we used hierarchical clustering to identify categories of ideas, the proportion of ideas that did not fall cleanly into any category, and the relative sizes of clusters.

Figure 2: Individual-Level Idea Quality Across Conditions



Note: Mean ratings for each submitter are displayed as jittered points, with condition means overlaid as larger points. Error bars are bootstrapped 95% confidence intervals.

3 Results

3.1 Individual-Level Quality

To examine the creativity of the submissions, we conducted separate mixed-effects models to predict each of the individual-level quality outcomes (overall creativity, originality and usefulness ratings from third-party evaluators) from the fixed effect of condition, with random effects for submitters and objects.

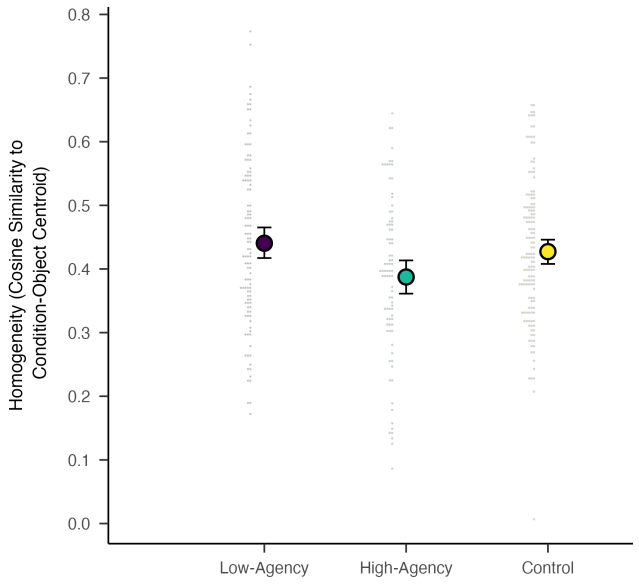
Contrary to our first hypothesis, there were no significant differences in overall idea creativity or originality across the AI-access conditions relative to control, $ps > .60$, as shown in Figure 2.

3.2 Group-Level Homogeneity

Our second hypothesis predicted that participants in the high-agency condition would have lower levels of homogenization across their submitted ideas than participants in the low-agency condition. We operationalized group-level homogeneity using the most common approach from prior research: the cosine similarity of each idea to the centroid embedding of the other ideas in the same condition for the same object [4]. Specifically, for each participant we randomly sampled one idea they had generated for each object (brick, bubble wrap, frisbee) and computed cosine similarity to the average embedding of all ideas other than the one being compared.

A linear mixed-effects model with condition as fixed effect and participants and objects as random effects revealed a significant effect of condition, $F(2, 119.72) = 4.73$, $p < .05$, $\eta_p^2 = .07$. Pairwise comparisons with Tukey's HSD revealed that participants in the high-agency condition submitted ideas that were less similar to the condition-object centroid than participants in the low-agency condition, indicated by significantly lower cosine similarity scores, $t(125) = -2.45$, $p < .05$. In contrast, there were no differences in idea homogeneity between the high-agency condition and the control condition, $t(122) = 1.87$, $p = .15$. This pattern of results suggests that while participants in the low-agency and control conditions produced ideas that were similarly homogeneous relative to the "average" idea in their respective conditions, the high-agency approach buffered against homogenization.

Figure 3: Similarity to Centroid Embeddings Across Conditions



Note: Each colored point represents the condition-level mean cosine similarity to the condition-object centroid embedding. Errors bars indicate bootstrapped 95% confidence intervals. Light gray points indicate the distribution of cosine similarity scores across each condition.

Although the centroid approach is standard among the literature on AI-induced homogenization in creative domains, it relies on the assumption that the structure of ideas can be distilled into a single point estimate of an average. However, as this approach aggregates across embeddings, it fails to capture the inherent variability in the structure of each condition's collective ideas. To address this concern, we conducted an exploratory clustering analysis (HDBSCAN [16]) to uncover categories of ideas, ideas that did not clearly fall into a single category of ideas, and evidence of participants glomming onto a single category—our key indicator of homogeneity at the group level.

We used three metrics to quantify the results of the clustering analysis: (1) number of unique clusters, (2) noise ratio, and (3) coefficient of variation. The number of unique clusters served as a proxy for the number of unique categories of uses proposed by participants in each condition. Noise ratio is the proportion of ideas that did not fall under any cluster, serving as a proxy for the "outlier" ideas indicative of greater diversity. The coefficient of variation (CV; standard deviation of cluster sizes divided by the mean cluster size) indicates the extent to which the clusters are equal in size; a higher CV suggests greater imbalance in cluster sizes. As shown in Figure 4, the three conditions showed distinct patterns in these metrics.

To test whether the observed differences in clustering patterns were statistically significant, we conducted permutation tests (n = 5,000 permutations) comparing the maximum and minimum condition means for each metric. For the number of unique clusters, the observed difference between conditions was 3.33, a difference driven by the statistically higher number of clusters in the control condition relative to the low-agency condition, $p = .04$. For noise ratio, the observed difference was 0.22, again driven by a higher noise ratio in the control condition relative to the low-agency, condition$p = .02$. For coefficient of variation, the observed difference was 0.75, driven by the low-agency condition having a significantly higher coefficient of variation than the control condition, $p = .005$. Taken together, the clustering results suggest that our experimental manipulation resulted in homogeneity outcomes significantly different from what would be expected by chance.

4 Discussion

In the present study, we investigated the relationship between human agency and creativity at both the individual and group levels, contributing to current discussions about the distribution of agency between human creators and AI creativity support tools [19]. Although we observed no differences in individual-level creativity via subjective assessments, our computational group-level analysis revealed distinct patterns of clustering. Within our sample, the control condition—which generated ideas without AI assistance—produced the highest number of clusters, the greatest noise ratio, and the displayed the greatest balance in cluster sizes (lowest coefficient of variation). In contrast, participants in the low-agency condition tended to glom onto a single, dominant cluster, manifesting as a lower noise ratio and higher coefficient of variation. These indicators of homogenization were attenuated among the high-agency condition. Our novel clustering analysis of ideas advances upon prior research on AI-induced homogenization by highlighting the distinction between a group of people generating ideas with low human agency (outsourcing the creative process to AI) and a group of people generating ideas with complete agency over the output.

4.1 Limitations

Notably, a number of participants in the treatment conditions refused to use the AI tool to complete the task. In feedback after the experiment, some participants shared that using the AI tool would have made the task boring (e.g., "I didn't use AI so I could enjoy the exercise.") and others refused to use the tool on principle (e.g., "I value the inherent uniqueness of the human experience. I do not believe AI has that.") Such psychological reactance would be in line with prior research on the forced adoption of self-service technology [5] or the broader phenomenon of aversion to AI in conventionally human domains [2]. While this somewhat compromised the random assignment, it also revealed real reactions to AI and digital literacy guidance that may occur in the wild, especially for a context as central to human identity as creation [17].

4.2 Implications

AI systems have lowered the barrier to entry for many forms of creative work and expression. Our study suggests that without an agentic approach people who use large language models may unwittingly contribute to an algorithmic monoculture. Setting aside glaring issues with the usage of LLMs in factual domains, mass reliance on the same model may have a more insidious effect on the everyday users who count on creative output being just a prompt

Figure 4: Clusters Across Conditions and Objects



Note: Clusters are visualized in 2D space using t-distributed stochastic neighbor embedding (t-SNE). Each unique cluster is shown in its own color for each condition-object pair. Ideas that were classified as "noise" by the clustering algorithm appear in light gray.

away [21]. Our findings of homogeneity among people who approached the AI system with low human agency align with prior research documenting large-scale changes in the way humans have communicated post-ChatGPT, from the use of flowery language in scientific publications [11] to increased semantic similarity on online platforms [3]. We posit human agency as one route to preserving the idiosyncrasy of human ideas and call upon creativity researchers and practitioners to scaffold agency in the AI-assisted creative process.

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